

Sharp Stability Strata for Finite Fabius–Rvachev Deconvolution

Colliding scales, vanishing factors, and optimal recovery exponents

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Abstract

Consider an independent sum of a known smooth compactly supported random variable and at most r unknown centered uniform factors. The Rvachev up distribution is an important special background, connecting the problem directly to the infinite sinc products and exact arithmetic developed in the ProveIt repository. We prove a complete local classification of the optimal inverse Hölder exponent in total variation distance. At a configuration with positive squared-scale multiplicities $m_1, \dots, m_s$ and m_0 zero slots, the optimal exponent for two varying nearby configurations is

$$\frac{1}{M}, \quad M = \max(\{m_1, \dots, m_s\} \cup \{2m_0 : m_0 > 0\}).$$

The estimate uses only the first r even cumulants for its upper bound. Sharpness is proved by explicit Chebyshev level-set constructions whose low-order power sums cancel. A vanishing m -factor cluster has a nonzero, explicitly identified leading density difference of order t^{2m} .

A distinct theorem concerns comparison with the fixed base configuration: its optimal exponent is 1 at a simple positive configuration and 1/2 at every singular configuration, regardless of multiplicity. Thus pointwise stability at a singular point is substantially stronger than uniform pairwise stability near it. We also give exact moment reconstruction, deterministic optimal-recovery consequences, noise-safe feasible fitting, an elementary sampling upper bound, and a formalization roadmap. Classical cumulant identities, Newton identities, and polynomial-root perturbation are distinguished from the classification proved here. The package contains exact algebraic checks and high-precision Fourier diagnostics; it does not claim a Lean verification, a statistical minimax lower bound, or worldwide priority.

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1 The research question and the contribution boundary

1.1 From exact identifiability to quantitative recovery

A product of sinc functions remembers its convolution factors. Exact recovery, however, is not the same question as stable recovery from inexact data. If two unknown factors have almost the same scale, or if several factors are becoming invisible because their scales approach zero, arbitrarily small changes in the observed law can conceal much larger changes in its factorization. The purpose of this article is to determine the exact power law of that loss of information when the number of unknown factors is bounded.

The ProveIt Fabius development supplies a particularly natural setting. Its module `Generalize dRvachevIdentifiability.lean` recovers admissible dyadic exponent sequences from analytic orders at dyadic points and establishes injectivity of the corresponding entire products [4]. That theorem uses zero multiplicities, not a noise metric on probability laws. The present problem allows finitely many arbitrary nonnegative scales, places them above a known smooth background, and asks for optimal stability in total variation. These are different hypotheses and different conclusions; the result below is not advertised as a replacement for the repository’s dyadic theorem.

The finite box convolutions and the infinite up product also occur in the repository’s holonomic study [5]. The smooth compactly supported function and its probabilistic interpretation have a much older literature, including Arias de Reyna’s account [1]. We will establish directly all properties of the up background required by our proofs.

1.2 What is classical and what is proved here

The reduction from cumulants to power sums, and then to a polynomial by Newton’s identities, is classical algebra. Collision singularities in moment inversion, Vieta maps, and Prony systems are extensively studied; Batenkov and Yomdin give a relevant geometric treatment [2]. The n th-root scale in degree- n polynomial-root perturbation is also classical; see Schätzle [3]. Our root-matching lemma is deliberately proved below rather than imported with an unspecified constant or an ambiguous treatment of multiplicities.

The principal result developed here is the combined classification for the *full output-law metric*: positive collisions contribute their multiplicity, zero clusters contribute twice their multiplicity, and the largest contribution controls the optimal local pairwise exponent. The fixed-base exponent has a different classification. Explicit smooth-density sharpness constructions and an exact vanishing-cluster leading term make both statements two-sided results, not merely condition-number heuristics.

This is a focused research contribution, not a claim to have settled all infinite-product inverse problems or a named longstanding conjecture. The source comparison comprised selected repository documents and targeted code searches, not an exhaustive audit of every report. No claim of worldwide novelty is made. The mathematical claims labeled theorems have proofs in this article; those proofs have not been independently refereed or checked by a proof assistant.

1.3 A precise repository snapshot

The pinned repository tree used for the principal source comparison was

0e0368b216b302d5822890b48bd19f2337af4d04.

The identifiability module and the holonomic report were read at that tree. Some search-index results referred to a different indexed tree, so search hits were used for discovery rather than as evidence that all files had the same version. The accompanying `SOURCE_AUDIT.md` records the scope and bibliographic checks. No repository files were changed.

2 Model, metrics, and the principal theorems

2.1 The fixed-background finite-factor model

Throughout, $r \geq 1$ is a fixed integer. Let B be a known random variable, independent of independent variables $U_1, \dots, U_r$, each uniform on $[-1, 1]$. Assume initially that B has a probability density

$$g \in C_c^\infty(\mathbb{R}), \quad \text{supp } g \subseteq [-L, L]. \quad (2.1)$$

Symmetry of g is not required. Define the closed ordered parameter cone

$$\mathcal{S}_r = \{a \in \mathbb{R}^r : 0 \leq a_1 \leq \dots \leq a_r\}, \quad X_a = B + \sum_{i=1}^r a_i U_i, \quad \mu_a = \mathcal{L}(X_a). \quad (2.2)$$

Its density is

$$f_a(x) = \mathbb{E}g\left(x - \sum_i a_i U_i\right). \quad (2.3)$$

A zero coordinate contributes the deterministic value zero. Thus r is a *capacity*: the model contains all sums with at most r positive uniform factors, padded to length r . There is no attempt to identify how many additional formal zero factors someone may have written outside this fixed representation. Ordering removes permutation ambiguity. Restricting to nonnegative amplitudes removes the sign ambiguity of a centered uniform distribution.

For $a, b \in \mathcal{S}_r$, put

$$d(a, b) = \max_i |a_i - b_i|, \quad d_{\text{TV}}(\mu_a, \mu_b) = \frac{1}{2} \|f_a - f_b\|_{L^1}. \quad (2.4)$$

The first metric is the bottleneck matching distance on the two multisets. For real ordered lists, matching in increasing order minimizes the largest matching distance: whenever two matched pairs cross, uncrossing them cannot increase that largest distance. Repeating this operation proves the assertion.

We use squared amplitudes and their power sums:

$$p_i = a_i^2, \quad s_k(a) = \sum_{i=1}^r p_i^k, \quad \mathcal{E}_r(a, b) = \max_{1 \leq k \leq r} |s_k(a) - s_k(b)|. \quad (2.5)$$

The normalization of this finite-data error assumes a fixed unit of scale; rescaling the physical variables changes the constants in our estimates.

2.2 Collision multiplicities

Fix $a^\circ \in \mathcal{S}_r$. Its squared coordinates have distinct positive values $c_1 < \dots < c_s$, with respective multiplicities $m_1, \dots, m_s$. Let m_0 be the number of zero coordinates. Thus $m_0 + \sum_{j=1}^s m_j = r$. Define

$$M(a^\circ) = \max(\{m_j : 1 \leq j \leq s\} \cup \{2m_0 : m_0 > 0\}). \quad (2.6)$$

The set in this maximum is nonempty, including when all coordinates vanish.

Definition 2.1 (Pairwise versus anchored exponents). A pairwise exponent $\alpha > 0$ is valid near a° if some relative neighborhood $\mathcal{U} \subseteq \mathcal{S}_r$ and constant C satisfy

$$d(a, b) \leq C d_{\text{TV}}(\mu_a, \mu_b)^\alpha \quad (a, b \in \mathcal{U}).$$

An anchored exponent is valid at a° if the analogous estimate is required only with $b = a^\circ$. “Optimal” means the largest valid exponent.

Theorem 2.2 (Complete local pairwise classification). *Under (2.1), the optimal pairwise exponent near a° is $1/M(a^\circ)$. More precisely, there are a relative neighborhood $\mathcal{U}$ and positive constants C_1, C_2 such that*

$$d(a, b) \leq C_1 \mathcal{E}_r(a, b)^{1/M(a^\circ)} \leq C_2 d_{\text{TV}}(\mu_a, \mu_b)^{1/M(a^\circ)} \quad (a, b \in \mathcal{U}). \quad (2.7)$$

In every relative neighborhood of a° , a bound for $d(a, b)$ by a constant times either $\mathcal{E}_r(a, b)^\alpha$ or $d_{\text{TV}}(\mu_a, \mu_b)^\alpha$ fails for every $\alpha > 1/M(a^\circ)$.

Theorem 2.3 (Complete anchored classification). *The optimal anchored exponent is 1 when all coordinates of a° are positive and pairwise distinct. In every other case it is $1/2$. In particular, if there is a repeated positive scale or at least one zero slot, then*

$$d(a, a^\circ) \leq C d_{\text{TV}}(\mu_a, \mu_{a^\circ})^{1/2} \quad (2.8)$$

for all sufficiently nearby a , and no larger anchored exponent is valid.

Theorem 2.4 (Uniform bounds on bounded parameter sets). *For every $A > 0$, there is a constant C such that*

$$d(a, b) \leq C d_{\text{TV}}(\mu_a, \mu_b)^{1/(2r)} \quad \text{whenever } 0 \leq a_1 \leq \dots \leq a_r \leq A. \quad (2.9)$$

The exponent $1/(2r)$ is optimal on this set. On the smaller set $a_ \leq a_1 \leq \dots \leq a_r \leq A$, where $0 < a_* < A$, the optimal uniform exponent is $1/r$. If in addition the squared coordinates have a fixed positive minimum separation, the inverse is uniformly Lipschitz on that separated set.*

2.3 Interpretation and an example table

The exponent measures a loss of resolution. With error level ε in the observed law, a cluster of m positive factors can force amplitude uncertainty of order $\varepsilon^{1/m}$. A cluster of m factors disappearing at zero can force uncertainty of order $\varepsilon^{1/(2m)}$. These orders are attained by explicit pairs of probability densities, not by arbitrary coefficient vectors that might fail to correspond to real nonnegative scales.

Configuration with capacity $r = 3$	M	Pairwise	Anchored
Three distinct positive scales	1	1	1
A repeated positive pair and a different positive scale	2	$1/2$	$1/2$
Three equal positive scales	3	$1/3$	$1/2$
One zero and two distinct positive scales	2	$1/2$	$1/2$
Two zeros and one positive scale	4	$1/4$	$1/2$
Three zeros	6	$1/6$	$1/2$

For a mixed example with three equal positive scales and two zero slots, $M = \max(3, 4) = 4$. Vanishing factors, rather than the positive collision, control the pairwise exponent. Constants depend on the background, the base configuration, the neighborhood, and the capacity; the theorems do not suppress this dependence.

3 The Fabius–Rvachev background and exact cumulants

3.1 A self-contained construction of the required background

Let $V_1, V_2, \dots$ be independent uniform variables on $[-1, 1]$ and define

$$B_{\text{up}} = \sum_{j=1}^{\infty} 2^{-j} V_j. \quad (3.1)$$

The series converges absolutely on every outcome and lies in $[-1, 1]$. We use the radian convention $\text{sinc } t = \sin(t)/t$, continued by $\text{sinc } 0 = 1$. The characteristic function is

$$\phi_{\text{up}}(t) = \mathbb{E}e^{itB_{\text{up}}} = \prod_{j=1}^{\infty} \text{sinc}(2^{-j}t). \quad (3.2)$$

Independence gives the finite products, and bounded convergence gives their limit. For fixed N and nonzero real t ,

$$|\phi_{\text{up}}(t)| \leq \prod_{j=1}^N \min(1, 2^j/|t|) \leq 2^{N(N+1)/2} |t|^{-N}. \quad (3.3)$$

Consequently $t^k \phi_{\text{up}}(t)$ is integrable for every fixed k . Fourier inversion and differentiation under the integral give a C^∞ density up. The support restriction of the random series gives $\text{supp up} \subseteq [-1, 1]$. It is therefore a C_c^∞ probability density, exactly the regularity needed in (2.1). This is the normalization of the up distribution used in the cited repository report [5]; the construction above also fixes it independently of notation in any other source.

In the convention $\widehat{g}(\xi) = \int e^{-2\pi i \xi x} g(x) dx$, the same formula is $\widehat{\text{up}}(\xi) = \prod_{h=0}^{\infty} \text{sinc}(\pi \xi / 2^h)$. The shift of index and the factor 2π account for the apparent difference between the two product displays.

3.2 Uniform cumulants are nonzero in every even order

For a compactly supported real random variable Y , define its cumulants by

$$\log \mathbb{E}e^{zY} = \sum_{n \geq 1} \kappa_n(Y) \frac{z^n}{n!}$$

in a neighborhood of zero. The logarithm is well defined there because the moment generating function has value one at zero. Independence adds cumulants, and scaling by a multiplies the n th cumulant by a^n .

Lemma 3.1 (Uniform cumulants). *If U is uniform on $[-1, 1]$, then its odd cumulants vanish and*

$$\gamma_{2k} := \kappa_{2k}(U) = \frac{2^{2k} B_{2k}}{2k} = (-1)^{k+1} \frac{(2k)! \zeta(2k)}{k \pi^{2k}} \neq 0. \quad (3.4)$$

In particular, $\gamma_2 = 1/3$, $\gamma_4 = -2/15$, and $\gamma_6 = 16/63$.

Proof. Direct integration gives $\mathbb{E}e^{zU} = \sinh z/z$. Its logarithmic derivative is $\coth z - 1/z$. The Bernoulli generating identity $w/(e^w - 1) = \sum_{n \geq 0} B_n w^n/n!$ gives

$$\coth z - \frac{1}{z} = \sum_{k \geq 1} \frac{2^{2k} B_{2k}}{(2k)!} z^{2k-1}.$$

Integration with zero constant term and comparison with the cumulant expansion prove the first expression. Alternatively, the classical product $\sinh z/z = \prod_{n \geq 1} (1 + z^2/(\pi^2 n^2))$ gives

$$\log \frac{\sinh z}{z} = \sum_{k \geq 1} \frac{(-1)^{k+1} \zeta(2k)}{k \pi^{2k}} z^{2k} \quad (|z| < \pi),$$

by absolutely convergent logarithmic expansion. This proves the second expression and its nonvanishing, since $\zeta(2k) > 0$. $\square$

Proposition 3.2 (Power-sum decoder). *In the model (2.2),*

$$s_k(a) = \frac{\kappa_{2k}(\mu_a) - \kappa_{2k}(B)}{\gamma_{2k}}. \quad (3.5)$$

For the up background,

$$\kappa_{2k}(B_{\text{up}}) = \frac{\gamma_{2k}}{4^k - 1}, \quad s_k(a) = \frac{\kappa_{2k}(\mu_a)}{\gamma_{2k}} - \frac{1}{4^k - 1}. \quad (3.6)$$

Proof. Cumulant additivity and scaling give (3.5). For the finite prefix of (3.1), the $2k$ th cumulant is $\gamma_{2k} \sum_{j=1}^N 2^{-2kj}$. Its moments converge to those of B_{up} by uniform boundedness and almost sure convergence. Cumulants are polynomials in finitely many moments, so they also converge. Summing the geometric series gives (3.6). $\square$

In particular, $\text{Var}(B_{\text{up}}) = 1/9$. This normalization check is useful: confusing the up variance with the variance $1/3$ of a single uniform factor would give an incorrect decoder at its first step.

4 Exact reconstruction and quantitative algebraic reduction

4.1 Newton reconstruction

For $p_i = a_i^2$, write

$$P_a(z) = \prod_{i=1}^r (z - p_i) = z^r - e_1 z^{r-1} + e_2 z^{r-2} - \cdots + (-1)^r e_r. \quad (4.1)$$

The elementary symmetric functions are recovered recursively from the power sums:

$$e_0 = 1, \quad e_j = \frac{1}{j} \sum_{k=1}^j (-1)^{k-1} e_{j-k} s_k \quad (1 \leq j \leq r). \quad (4.2)$$

For completeness, define $E(t) = \prod_i (1 + p_i t) = \sum_j e_j t^j$. Logarithmic differentiation gives

$$E'(t) = E(t) \sum_{k \geq 1} (-1)^{k-1} s_k t^{k-1}.$$

Equating coefficients proves (4.2).

Proposition 4.1 (Finite exact identifiability). *The first r even cumulants of μ_a , together with the known background cumulants, determine $a \in \mathcal{S}_r$ uniquely. The same is true of the raw moments through order $2r$. The reconstruction is: decode $s_1, \dots, s_r$, compute (4.2), form (4.1), list its roots with multiplicity, and take their nonnegative square roots in increasing order.*

Proof. Proposition 3.2 supplies the power sums. Newton's identities supply all coefficients of P_a . A monic polynomial determines its complex root multiset, and in this model all those roots are real and nonnegative. Nonnegative square root and ordering therefore determine a . Cumulants through order $2r$ are polynomials in the raw moments through that order, proving the final assertion. $\square$

No division by the observed characteristic function is used. In particular, zeros of the known up characteristic function cause no obstruction to this exact finite-dimensional decoder.

4.2 Total variation controls the required finite data

Suppose temporarily that $a_r, b_r \leq A$ and set $R = \max(1, L + rA)$. All observed laws are supported on $[-R, R]$. If $\varepsilon = d_{\text{TV}}(\mu_a, \mu_b)$, then their raw moments satisfy

$$|m_n(\mu_a) - m_n(\mu_b)| \leq 2R^n \varepsilon. \quad (4.3)$$

Lemma 4.2 (An explicit cumulant bound). *Let $F_n = \sum_{q=1}^n q! \left\{ \begin{smallmatrix} n \\ q \end{smallmatrix} \right\}$, where $\left\{ \begin{smallmatrix} n \\ q \end{smallmatrix} \right\}$ is a Stirling number of the second kind. Then*

$$|\kappa_n(\mu_a) - \kappa_n(\mu_b)| \leq 2R^n F_n \varepsilon. \quad (4.4)$$

Consequently

$$\mathcal{E}_r(a, b) \leq C_{r,R} \varepsilon, \quad C_{r,R} = \max_{1 \leq k \leq r} \frac{2R^{2k} F_{2k}}{|\gamma_{2k}|}. \quad (4.5)$$

The same bound holds through any other fixed finite cumulant order.

Proof. The moment–cumulant formula is

$$\kappa_n(\mu) = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{D \in \pi} m_{|D|}(\mu),$$

where Π_n is the set of partitions of $\{1, \dots, n\}$. For a partition with q blocks, telescope the difference of the two products. Each of its q terms is bounded by $2R^n \varepsilon$, by (4.3) and $|m_j| \leq R^j$. Multiplication by $(q-1)!$ and summation over the $\left\{ \begin{smallmatrix} n \\ q \end{smallmatrix} \right\}$ partitions gives (4.4). The known background cancels in differences in (3.5); division by the nonzero γ_{2k} gives (4.5). $\square$

These constants are explicit and conservative, not asserted to be numerically optimal. Their purpose is to make every finite-order continuity assertion quantitative.

4.3 Power-sum error controls polynomial-coefficient error

For monic degree- r polynomials, let $\|P - Q\|_{\text{coef}}$ be the largest absolute difference between their nonleading coefficients. Put $H_0 = A^2$. If $\delta = \mathcal{E}_r(a, b)$, then

$$\|P_a - P_b\|_{\text{coef}} \leq K_{r,A} \delta. \quad (4.6)$$

One explicit choice is $K_{r,A} = \max_{1 \leq j \leq r} K_j$, where

$$K_0 = 0, \quad K_j = \frac{1}{j} \sum_{k=1}^j \left(r H_0^k K_{j-k} + \binom{r}{j-k} H_0^{j-k} \right). \quad (4.7)$$

Indeed, $|s_k| \leq r H_0^k$ and $|e_j| \leq \binom{r}{j} H_0^j$. Subtracting the two Newton recurrences, telescoping each product $e_{j-k} s_k$, and inducting on j proves (4.6) with exactly (4.7).

4.4 The regular locus and its Jacobian

The power-sum map in squared coordinates has Jacobian

$$\frac{\partial s_k}{\partial p_i} = k p_i^{k-1}, \quad \det \left(\frac{\partial s_k}{\partial p_i} \right)_{k,i=1}^r = r! \prod_{i < j} (p_j - p_i). \quad (4.8)$$

The determinant follows by factoring k from row k and using the Vandermonde determinant. Passing to amplitude coordinates gives

$$\det \left(\frac{\partial s_k}{\partial a_i} \right)_{k,i=1}^r = 2^r r! \left(\prod_i a_i \right) \prod_{i < j} (a_j^2 - a_i^2). \quad (4.9)$$

Thus the regular positive locus is precisely the locus of distinct positive scales. The determinant detects where local Lipschitz inversion can fail, but by itself does not give the optimal exponent on the singular locus. That is the additional content of the main theorems.

5 Root matching with multiplicities and separated cluster factors

5.1 A quantitative root-matching lemma

A bound saying that each root is close to *some* root of the other polynomial is insufficient: it can ignore multiplicities. We need a matching that counts every root.

Lemma 5.1 (Root matching on a bounded real interval). *Let P, Q be monic degree- n polynomials with all roots in $[0, H_0]$. There is a constant C_{n,H_0} such that their ordered root lists p, q satisfy*

$$\max_i |p_i - q_i| \leq C_{n,H_0} \|P - Q\|_{\text{coef}}^{1/n}. \quad (5.1)$$

For sufficiently small $\eta = \|P - Q\|_{\text{coef}} > 0$, one may take $C_{n,H_0} = 4n(2H)^{1/n}$, with $H = \sum_{\ell=0}^{n-1} (H_0 + 1)^\ell$.

Proof. For $|z| \leq H_0 + 1$, $|P(z) - Q(z)| \leq H\eta$. Set $\rho = (2H\eta)^{1/n}$ and suppose $\rho < 1/2$. Take the union of the disks of radius ρ' centered at the roots of P , where $\rho \leq \rho' < 2\rho$ is chosen to avoid tangencies of boundary circles. All boundary points satisfy $|z| \leq H_0 + 1$ and have distance at least ρ' from every root of P . Hence on that boundary,

$$|P(z)| = \prod_i |z - p_i| \geq (\rho')^n \geq 2H\eta > |P(z) - Q(z)|.$$

Rouché's theorem, applied to the oriented boundary of each component, gives the same number of zeros of P and Q in that component, with multiplicity. Piecewise circular boundaries can equivalently be perturbed slightly before applying the argument principle. The total count is n , so every root of Q is in one of these components.

A component contains at most n disks and its disk-intersection graph is connected. Joining centers along this graph shows that its diameter is at most $2n\rho' < 4n\rho$. Match equal numbers of roots within each component. This gives a full matching with displacement at most $4n\rho$. Sorting the two real lists cannot worsen the largest displacement, proving the displayed bound for small η . If $\eta = 0$, the polynomials coincide. For the remaining η , enlarge the constant using the trivial bound $\max_i |p_i - q_i| \leq H_0$ and the fixed positive threshold for η . $\square$

5.2 Separated clusters depend Lipschitz-continuously on the coefficients

Lemma 5.2 (Lipschitz extraction of an isolated factor). *Let P° be monic with distinct root clusters separated by disjoint circles $\Gamma_0, \dots, \Gamma_s$. For nearby monic polynomials whose roots remain inside the corresponding circles with the same cluster counts, the monic factor belonging to any one circle is Lipschitz in the full coefficient vector. Its degree is the number of roots in that circle, counted with multiplicity.*

Proof. For a circle Γ_j and $k \geq 1$, the cluster power sum is

$$t_{j,k}(P) = \frac{1}{2\pi i} \oint_{\Gamma_j} z^k \frac{P'(z)}{P(z)} dz. \quad (5.2)$$

The residue at a root of multiplicity m is m times the corresponding k th power, so the formula includes all multiplicities. Shrink the coefficient neighborhood so that $|P|$ is uniformly bounded below on the fixed circles. The numerators and derivatives are uniformly bounded there. The identity

$$\frac{P'}{P} - \frac{Q'}{Q} = \frac{P'Q - Q'P}{PQ}$$

then bounds the integrand difference by a constant times $\|P - Q\|_{\text{coef}}$. Integration gives a Lipschitz bound for the finitely many cluster sums $t_{j,1}, \dots, t_{j,m_j}$. Applying the finite Newton recurrences to these sums gives the coefficients of the monic degree- m_j factor and preserves Lipschitz dependence on this bounded neighborhood. $\square$

The use of a fixed contour is important. The roots *within* a cluster may collide, but the cluster as a whole remains separated from the other clusters. We are not assuming differentiable or Lipschitz individual roots at a collision.

6 Upper bounds: proof of the pairwise and global estimates

Proof of the upper bound in Theorem 2.2. Choose small disjoint disks around 0 when $m_0 > 0$ and around each positive c_j . Choose a relative parameter neighborhood $\mathcal{U}$ of a° so that exactly the prescribed coordinate block stays in each disk. All parameters in this neighborhood are bounded by some A .

Let $\delta = \mathcal{E}_r(a, b)$. By (4.6), the coefficient vectors of P_a and P_b differ by at most a constant times δ . Lemma 5.2 gives the same type of estimate for the monic factor belonging to cluster j . Applying Lemma 5.1 to that degree- m_j factor yields

$$\max_{i \text{ in cluster } j} |a_i^2 - b_i^2| \leq C_j \delta^{1/m_j}. \quad (6.1)$$

Near a positive center c_j , both square roots are bounded away from zero. The identity

$$|a_i - b_i| = \frac{|a_i^2 - b_i^2|}{a_i + b_i}$$

therefore preserves the exponent $1/m_j$.

For the zero cluster, the same argument in squared coordinates gives exponent $1/m_0$. For nonnegative u, v ,

$$|\sqrt{u} - \sqrt{v}| \leq \sqrt{|u - v|}, \quad (6.2)$$

so that cluster gives amplitude exponent $1/(2m_0)$. Taking the maximum of these bounds, and first treating $0 \leq \delta \leq 1$, gives exponent $1/M(a^\circ)$. An enlargement of the constant handles the remaining bounded range of δ . Finally, Lemma 4.2 gives $\delta \leq C_{r,R} d_{\text{TV}}(\mu_a, \mu_b)$. $\square$

Proof of the upper estimates in Theorem 2.4. On the full bounded parameter set, (4.6) followed by Lemma 5.1 in degree r yields squared-scale exponent $1/r$. Applying (6.2) gives amplitude exponent $1/(2r)$, and (4.5) converts to total variation. If all amplitudes are at least $a_* > 0$, the denominator $a_i + b_i \geq 2a_*$ preserves exponent $1/r$ instead.

For a uniform squared-scale separation $\Delta > 0$, sufficiently small coefficient perturbations place each matched root q_i within $\Delta/2$ of p_i . This follows uniformly from Lemma 5.1. Since $Q(q_i) = 0$,

$$|P(q_i)| = |P(q_i) - Q(q_i)| \leq H_1 \|P - Q\|_{\text{coef}}$$

for a fixed H_1 on the bounded interval. Every factor $|q_i - p_j|$, $j \neq i$, is at least $\Delta/2$, giving

$$|q_i - p_i| \leq H_1 (2/\Delta)^{r-1} \|P - Q\|_{\text{coef}}. \quad (6.3)$$

Take square roots using the positive lower bound a_* , then use (4.6) and (4.5). Large coefficient perturbations are covered by enlarging the constant, using the bounded parameter diameter and the fixed threshold separating the small and large cases. For $r = 1$, the empty product in (6.3) is one and no separation condition is needed. $\square$

Remark 6.1 (How much background regularity is used so far?). The upper estimates use only that the known background law has bounded support. They do not require a density or smoothness. Smoothness becomes essential to the sharpness constructions below, where high-order cancellation must persist in the strong L^1 metric. Thus the stated classification is a theorem for the smooth-background model, while its upper half applies more widely.

7 Explicit moment-canceling clusters

The sharpness proof must stay inside the real nonnegative scale model. Generic perturbations of a polynomial with a repeated root often produce complex roots, so they would not suffice. Chebyshev level sets give suitable real configurations at every multiplicity.

Lemma 7.1 (Two real multisets with maximal initial cancellation). *For $m \geq 1$, let $x_1^+ < \dots < x_m^+$ and $x_1^- < \dots < x_m^-$ be the roots of $T_m(x) = 1/2$ and $T_m(x) = -1/2$, respectively. Both lists consist of simple roots in $(-1, 1)$ and have no root in common. Their power sums satisfy*

$$\sum_i (x_i^+)^k = \sum_i (x_i^-)^k \quad (1 \leq k < m), \quad (7.1)$$

$$D_m := \sum_i (x_i^+)^m - \sum_i (x_i^-)^m = m 2^{1-m} > 0. \quad (7.2)$$

The shifted lists $u_i^\pm = 2 + x_i^\pm \in (1, 3)$ obey the same initial equalities and the same m th-power difference D_m .

Proof. The identity $T_m(\cos \theta) = \cos(m\theta)$ shows that each horizontal level in $(-1, 1)$ meets the graph in exactly m simple points of $(-1, 1)$. Simplicity also follows because at such a point $\sin(m\theta) \neq 0$ and $T'_m(\cos \theta) = m \sin(m\theta) / \sin \theta \neq 0$.

The monic polynomials of the two root lists are

$$Q_\pm(z) = 2^{1-m} (T_m(z) \mp 1/2). \quad (7.3)$$

They have identical nonconstant coefficients. The constant coefficient of Q_+ minus that of Q_- equals -2^{1-m} . Newton's recurrence in coefficient form says, for $1 \leq k \leq m$,

$$s_k + c_1 s_{k-1} + \dots + c_{k-1} s_1 + k c_k = 0,$$

where c_j is the coefficient of z^{m-j} . Thus the sums agree for $k < m$, and at $k = m$ their difference is $-m(-2^{1-m}) = D_m$. The root sets are disjoint because a number cannot simultaneously satisfy $T_m(x) = 1/2$ and $T_m(x) = -1/2$. Finally, expanding $(2 + x)^k$ binomially proves both assertions for $u_i^\pm$. $\square$

Example 7.2 (A triple collision). For $m = 3$, the two monic polynomials are

$$Q_+(z) = z^3 - \frac{3}{4}z - \frac{1}{8}, \quad Q_-(z) = z^3 - \frac{3}{4}z + \frac{1}{8}.$$

Both root lists have sum zero and sum of squares $3/2$, but their sums of cubes differ by $3/4$. This produces two nearby three-factor configurations with exactly the same second and fourth output cumulants, while the first differing cumulant is sixth order. A perturbation near a positive squared scale makes that difference order t^3 ; a perturbation from zero in amplitude coordinates makes it order t^6 .

8 Smoothness, cancellation, and leading density terms

8.1 Translation Taylor expansion in L^1

Lemma 8.1 (A uniform translation expansion). *Let $k \in C_c^\infty(\mathbb{R})$ and let Z be a bounded random variable. For every integer $n \geq 1$,*

$$\mathbb{E}k(\cdot - tZ) = \sum_{\ell=0}^n \frac{(-t)^\ell \mathbb{E}Z^\ell}{\ell!} k^{(\ell)} + o_{L^1}(t^n) \quad (t \rightarrow 0). \quad (8.1)$$

If two bounded variables have identical moments through order $n - 1$, the difference of their expansions starts with their n th-moment difference.

Proof. The integral Taylor formula for translation is

$$k(\cdot - h) - \sum_{\ell=0}^{n-1} \frac{(-h)^\ell}{\ell!} k^{(\ell)} = \frac{(-h)^n}{(n-1)!} \int_0^1 (1-v)^{n-1} k^{(n)}(\cdot - vh) dv.$$

Translations are continuous in L^1 . Replacing the translated $k^{(n)}$ in the integral by $k^{(n)}$ therefore changes the expression by $o_{L^1}(|h|^n)$, uniformly for $|h|$ in a shrinking interval. Substitute $h = tZ$. Boundedness of Z makes the remainder uniform over its possible values; taking expectations proves (8.1). $\square$

Convolving g with any fixed collection of finite uniform factors gives another smooth compactly supported density k , with $\|k^{(n)}\|_1 \leq \|g^{(n)}\|_1$. Moreover,

$$\|k^{(n)}\|_1 > 0 \quad (n \geq 1). \quad (8.2)$$

Otherwise $k^{(n)} = 0$ everywhere by continuity, so k is a polynomial of degree less than n on $\mathbb{R}$. A compactly supported polynomial vanishes identically, contradicting $\int k = 1$.

8.2 Smooth symmetric dependence on positive squared scales

Lemma 8.2 (Positive-cluster Taylor cancellation). *For a fixed smooth compactly supported density k and a fixed $c > 0$, define*

$$\mathcal{F}(p_1, \dots, p_m)(x) = \mathbb{E}k\left(x - \sum_{i=1}^m \sqrt{p_i} U_i\right) \quad (p_i > 0).$$

This is a symmetric C^∞ map into $L^1(\mathbb{R})$. If two direction vectors $v, w \in \mathbb{R}^m$ have the same power sums through order $m - 1$, then

$$\mathcal{F}(c\mathbf{1} + tv) - \mathcal{F}(c\mathbf{1} + tw) = t^m H + o_{L^1}(t^m) \quad (8.3)$$

for some $H \in L^1$.

Proof. On a compact subset of the positive orthant, every derivative of $\sqrt{p}$ is bounded. Differentiating the displayed expectation produces finite sums of bounded functions of the U_i times translated derivatives $k^{(j)}$. Their L^1 norms have uniform integrable bounds. Translation continuity and differentiation under the expectation give C^∞ dependence into L^1 . Permutation symmetry follows from the identical independent laws of the U_i .

Expand the map at $c\mathbf{1}$ through total degree m . Its homogeneous term of degree ℓ is a symmetric polynomial in the direction coordinates, with coefficients in L^1 . Every symmetric polynomial of degree ℓ is a polynomial in the elementary symmetric functions $e_1, \dots, e_\ell$ with their usual weights, hence in the power sums $s_1, \dots, s_\ell$ by Newton's identities. This algebraic statement also holds for vector-valued coefficients: their span in this finite Taylor polynomial is finite dimensional, and one applies the scalar identity in a basis of that span. One elementary proof of the scalar statement successively subtracts the product $e_1^{\lambda_1 - \lambda_2} e_2^{\lambda_2 - \lambda_3} \dots e_m^{\lambda_m}$ corresponding to the leading ordered monomial $z_1^{\lambda_1} \dots z_m^{\lambda_m}$; the leading monomial decreases until none remains. The weighted degree is preserved throughout.

Consequently all Taylor terms of degrees below m agree for v and w . The difference of their degree- m terms is H , and the Banach-space Taylor remainder is $o_{L^1}(t^m)$, proving (8.3). $\square$

9 Sharpness at positive collisions

Fix a positive squared-scale cluster of size m at $c > 0$, and hold all other coordinates equal to their values at a° . Let k be the density of the known background convolved with these fixed factors. For the Chebyshev lists from Lemma 7.1, replace the cluster by

$$p_i^\pm(t) = c + tx_i^\pm, \quad a_i^\pm(t) = \sqrt{c + tx_i^\pm}, \quad t > 0. \quad (9.1)$$

For small t , all scales remain positive and in their original isolated cluster. The full configurations converge to a° and satisfy

$$d(a^+(t), a^-(t)) \sim \frac{t}{2\sqrt{c}} \max_i |x_i^+ - x_i^-|. \quad (9.2)$$

The coefficient on the right is positive because the two ordered root lists differ.

By binomial expansion and Lemma 7.1, the cluster power sums agree through order $m - 1$, while their order- m difference is exactly $D_m t^m$. The fixed coordinates do not change those differences. Therefore

$$\kappa_{2m}(\mu_{a^+(t)}) - \kappa_{2m}(\mu_{a^-(t)}) = \gamma_{2m} D_m t^m \neq 0, \quad (9.3)$$

and all lower-order cumulants of the output laws agree. Odd cumulants, even when the background is not symmetric, agree because the uniform factors have zero odd cumulants and the background is the same.

Proposition 9.1 (Positive-collision sharpness in total variation). *For the paths (9.1), there is a nonzero compactly supported $H_{c,m,k} \in L^1(\mathbb{R})$ such that*

$$f_{a^+(t)} - f_{a^-(t)} = t^m H_{c,m,k} + o_{L^1}(t^m), \quad (9.4)$$

$$d_{\text{TV}}(\mu_{a^+(t)}, \mu_{a^-(t)}) \sim \frac{1}{2} \|H_{c,m,k}\|_1 t^m. \quad (9.5)$$

In addition, $\mathcal{E}_r(a^+(t), a^-(t)) \asymp t^m$.

Proof. Lemma 8.2 gives the first expansion. All densities along these paths have support in one fixed compact interval; the Taylor coefficients, being derivatives of such densities, have support there as well. Since all lower cumulants agree, the moment–cumulant polynomial implies that the raw moments

of order $2m$ differ by the cumulant difference (9.3). Dividing the L^1 expansion by t^m and integrating against x^{2m} on that fixed compact interval gives

$$\int x^{2m} H_{c,m,k}(x) dx = \gamma_{2m} D_m \neq 0.$$

Thus $H_{c,m,k}$ is nonzero. Taking L^1 norms proves (9.5). Finally, the m th power-sum gap is $D_m t^m$, while for every fixed $k \geq m$ the binomial expansion makes its gap $O(t^m)$. As $m \leq r$, the maximum over $1 \leq k \leq r$ is bounded above and below by positive constants times t^m . $\square$

By (9.2) and (9.5), a putative pairwise bound with exponent $\alpha > 1/m$ would require $t \leq C' t^{m\alpha}$ along these paths, which is impossible as $t \downarrow 0$. This proves the positive-cluster sharpness half of Theorem 2.2. The argument includes $m = 1$ and therefore also proves that a regular exponent larger than one cannot hold.

10 Vanishing factors: an exact leading law and doubled exponents

The zero cluster has an extra degeneracy: the law of aU is even as a function of the amplitude a , so its first detectable change occurs in a^2 , not a . The next theorem quantifies how this interacts with an m -fold collision.

Theorem 10.1 (Exact vanishing-cluster leading term). *Suppose a° has a zero cluster of size $m \geq 1$. Hold all other scales fixed, and let k be the smooth compactly supported density of the background plus those fixed factors. Set $u_i^\pm = 2 + x_i^\pm$ using Lemma 7.1, and turn the zero cluster into*

$$a_i^\pm(t) = t\sqrt{u_i^\pm}, \quad t > 0. \quad (10.1)$$

Then

$$f_{a^+(t)} - f_{a^-(t)} = \frac{\gamma_{2m} D_m}{(2m)!} t^{2m} k^{(2m)} + o_{L^1}(t^{2m}), \quad (10.2)$$

and in particular

$$d_{\text{TV}}(\mu_{a^+(t)}, \mu_{a^-(t)}) \sim \frac{|\gamma_{2m} D_m|}{2(2m)!} \|k^{(2m)}\|_1 t^{2m}. \quad (10.3)$$

The constant is strictly positive. Meanwhile,

$$d(a^+(t), a^-(t)) = t \max_i |\sqrt{u_i^+} - \sqrt{u_i^-}| > 0, \quad (10.4)$$

and $\mathcal{E}_r(a^+(t), a^-(t)) \asymp t^{2m}$.

Proof. Define bounded symmetric random variables $Z_\pm = \sum_i \sqrt{u_i^\pm} U_i$. Their even cumulants are $\gamma_{2j} \sum_i (u_i^\pm)^j$. By Lemma 7.1, these agree for $j < m$, and their order- $2m$ difference is $\gamma_{2m} D_m$. Odd cumulants vanish. The moment–cumulant identities therefore give equal raw moments through order $2m - 1$ and

$$\mathbb{E} Z_+^{2m} - \mathbb{E} Z_-^{2m} = \gamma_{2m} D_m.$$

The two output densities are $\mathbb{E} k(\cdot - tZ_+)$ and $\mathbb{E} k(\cdot - tZ_-)$. Apply Lemma 8.1 with $n = 2m$ and subtract. The sign $(-1)^{2m}$ is positive, giving exactly (10.2).

Both γ_{2m} and D_m are nonzero, and $\|k^{(2m)}\|_1 > 0$ by (8.2). Norm convergence gives (10.3). The amplitude formula is immediate. In squared coordinates $p_i^\pm = t^2 u_i^\pm$, the sums agree through order $m - 1$ and differ at order m by $D_m t^{2m}$. Higher fixed orders have differences $O(t^{2m+2})$ or smaller, proving the final assertion. $\square$

For $m = 3$, the signed coefficient in (10.2) is

$$\frac{(16/63)(3/4)}{6!} = \frac{1}{3780}.$$

Thus a vanishing triple has density difference $t^6 k^{(6)}/3780 + o_{L^1}(t^6)$ and total variation asymptotic $t^6 \|k^{(6)}\|_1/7560$. The amplitude separation is proportional to t . This is an explicit sixth-order obstruction, not merely a qualitative non-Lipschitz example.

Completion of Theorems 2.2 and 2.4. Choose a cluster attaining $M(a^\circ)$. If it is positive, use Proposition 9.1; if it is zero, use Theorem 10.1. In either case the two configurations tend to a° , their amplitude separation is comparable to t , and both their finite-data error and their total variation are comparable to $t^{M(a^\circ)}$. Every neighborhood therefore contains pairs excluding every exponent greater than $1/M(a^\circ)$. Together with Section 6, this proves Theorem 2.2 in full.

For the full box $0 \leq a_i \leq A$, use a zero cluster of size r at the origin. For the positive box $a_* \leq a_i \leq A$, choose all coordinates equal to any value strictly between a_* and A , and use a positive cluster of size r . These constructions remain inside their boxes for small t and prove the claimed optimal global exponents in Theorem 2.4. $\square$

11 Anchored recovery and why its exponent is different

11.1 A nonnegative polynomial certificate

The poor pairwise examples compare two moving, highly correlated clusters. When one configuration is held exactly at the singular base, a nonnegative polynomial blocks this cancellation.

Upper bound in Theorem 2.3. The simple positive case follows from Theorem 2.2. For the other cases, let $c_1, \dots, c_s$ be the distinct positive squared centers of the base. Define

$$Q(x) = \begin{cases} \prod_{j=1}^s (x - c_j)^2, & m_0 = 0, \\ x \prod_{j=1}^s (x - c_j)^2, & m_0 > 0. \end{cases} \quad (11.1)$$

The empty product is one, so if all coordinates vanish then $Q(x) = x$. This polynomial is nonnegative for $x \geq 0$ and vanishes at every base squared coordinate. If $Q(x) = \sum_{\ell=0}^d q_\ell x^\ell$, then

$$0 \leq \sum_i Q(a_i^2) = \sum_{\ell=1}^d q_\ell (s_\ell(a) - s_\ell(a^\circ)) \leq C d_{\text{TV}}(\mu_a, \mu_{a^\circ}). \quad (11.2)$$

The constant terms cancel because both lists have length r . The last inequality uses the decoder and Lemma 4.2 through the fixed finite order $2d$; it does not assume $d \leq r$.

Near a positive center c_j , the ratio $Q(x)/(x - c_j)^2$ is continuous and strictly positive. Hence $Q(x) \geq b_j(x - c_j)^2$ there for some $b_j > 0$. When there is a zero cluster, $Q(x)/x = \prod_j (x - c_j)^2$ is strictly positive near zero, so $Q(x) \geq b_0 x$ for small $x \geq 0$. Shrink the parameter neighborhood so its coordinate blocks stay near their prescribed centers. Every summand in (11.2) is nonnegative. It follows that

$$|a_i^2 - c_j| \leq C\sqrt{\varepsilon} \quad \text{in a positive block,} \quad a_i^2 \leq C\varepsilon \quad \text{in the zero block,}$$

where $\varepsilon = d_{\text{TV}}(\mu_a, \mu_{a^\circ})$. The positive-block square-root map is Lipschitz, while in the zero block taking square roots gives $a_i \leq C\sqrt{\varepsilon}$. Combining the blocks proves (2.8). $\square$

11.2 Optimality of the square-root anchored exponent

Sharpness and completion of Theorem 2.3. If a zero coordinate is present, change just that amplitude from zero to t , leaving all other coordinates fixed and sorting afterward. Let k be the density of the background and those fixed factors. Since $\mathbb{E}U = 0$ and $\mathbb{E}U^2 = 1/3$, Lemma 8.1 gives

$$f_t - k = \frac{t^2}{6}k'' + o_{L^1}(t^2), \quad d_{\text{TV}}(\mu_t, \mu_{a^\circ}) \sim \frac{\|k''\|_1}{12}t^2. \quad (11.3)$$

The parameter distance is t . The positive leading constant excludes every anchored exponent greater than $1/2$.

If there is no zero but a repeated positive squared scale c , replace two copies of c by $c + t$ and $c - t$. The first cluster power sum stays equal to $2c$, while the second increases by $2t^2$. Positive-coordinate smoothness shows that the density has an L^1 Taylor expansion. Its linear term cancels by symmetry in the two chosen coordinates, so the difference from the base is $t^2H + o_{L^1}(t^2)$. The output fourth cumulant changes by $2\gamma_4t^2 \neq 0$, whereas all lower cumulants agree. Integrating the expansion against x^4 on a common compact support gives $\int x^4H = 2\gamma_4 \neq 0$. Thus total variation is comparable to t^2 , while amplitude distance is comparable to t . Again an exponent greater than $1/2$ is impossible.

Finally, at a simple positive base, change one squared scale from c to $c + t$. Smooth dependence gives an $O(t)$ upper bound in total variation, and the variance change γ_2t gives a matching lower bound by Lemma 4.2. Parameter distance is comparable to t , excluding anchored exponents greater than one. The previously proved upper bounds complete all cases. $\square$

Remark 11.1 (There is no contradiction between the two classifications). At an m -fold positive collision, each Chebyshev configuration is generally at law distance of order t^2 from the singular base, while their distance from each other can be of order t^m . At a zero cluster, each configuration has a variance change of order t^2 from the base, even though the two output laws can differ only at order t^{2m} . Anchoring removes the possibility that their common low-order changes cancel against one another.

12 Deterministic recovery, feasible fitting, and sampling

12.1 A forward continuity bound

Using the same uniforms to couple two models and the translation bound $\|g(\cdot - u) - g(\cdot - v)\|_1 \leq \|g'\|_1 |u - v|$, we obtain

$$d_{\text{TV}}(\mu_a, \mu_b) \leq \frac{\|g'\|_1}{4} \sum_{i=1}^r |a_i - b_i| \leq \frac{r\|g'\|_1}{4} d(a, b). \quad (12.1)$$

Here $\mathbb{E}|U_i| = 1/2$ supplies the factor $1/4$ after the total-variation normalization. This bound ensures continuity of the forward map on compact parameter sets. It need not be sharp at singular points.

12.2 Optimal deterministic ambiguity radii

Take a compact relative neighborhood K of a° small enough that (2.7) holds. Suppose an observed probability law ν obeys $d_{\text{TV}}(\nu, \mu_a) \leq \varepsilon$ for an unknown $a \in K$. Minimize $d_{\text{TV}}(\nu, \mu_b)$ over $b \in K$; continuity and compactness ensure a minimizer $\hat{a}$. It satisfies

$$d_{\text{TV}}(\mu_{\hat{a}}, \mu_a) \leq 2\varepsilon, \quad d(\hat{a}, a) \leq C(2\varepsilon)^{1/M(a^\circ)}.$$

An exact minimizer is used to state the information-theoretic bound, not to claim an efficient numerical algorithm.

Conversely, take either sharpness pair $\mu_+(t), \mu_-(t)$ and let $\nu_t = (\mu_+(t) + \mu_-(t))/2$. Each model is at total-variation distance $d_{\text{TV}}(\mu_+(t), \mu_-(t))/2$ from ν_t . Every single output of an estimator on this common observation is at parameter distance at least $d(a^+(t), a^-(t))/2$ from one of the two models. With $t = c\varepsilon^{1/M}$ and a sufficiently small fixed c , both models lie in the allowed ε -ball for all sufficiently small ε . Therefore the worst-case deterministic recovery error is bounded below by a positive constant times $\varepsilon^{1/M}$. Together with the upper bound, this proves the optimal deterministic rate near the base.

12.3 Exact finite-data reconstruction is not automatically noise-safe

For exact data, the decoder in Proposition 4.1 is sufficient. Noisy data require an additional feasibility step. A polynomial formed by blindly applying Newton's identities to perturbed sums may have negative or nonreal roots, even when the perturbation is tiny. Taking real parts of its roots is not justified by the theorems in this article.

A mathematically safe alternative is the following finite-dimensional fit. Given approximate sums $\tilde{s}_1, \dots, \tilde{s}_r$ and a compact parameter set K , minimize

$$\rho(b) = \max_{1 \leq k \leq r} |s_k(b) - \tilde{s}_k| \quad \text{over } b \in K. \quad (12.2)$$

A minimizer exists. If the true parameter $a \in K$ has $\max_k |s_k(a) - \tilde{s}_k| \leq \eta$, then every minimizer $\hat{a}$ has

$$\mathcal{E}_r(\hat{a}, a) \leq 2\eta, \quad d(\hat{a}, a) \leq C(2\eta)^{1/M} \quad (12.3)$$

when K is a local neighborhood as above. On the global box, the corresponding exponent is $1/(2r)$; on a positive box it is $1/r$. An approximate minimizer with objective at most $\inf \rho + \tau$ gives $2\eta + \tau$ in place of 2η .

For certified computation one may work in squared coordinates $0 \leq p_1 \leq \dots \leq p_r \leq A^2$ and use interval subdivision or polynomial inequality methods to enclose the feasible set. This is a valid route to a certificate, but a fast implementation and tight complexity analysis are not provided here. The included executable code validates the exact algebraic stage, not a complete certified nonlinear optimizer.

12.4 A sampling upper bound, with its limits made explicit

The total variation distance between a continuous law and its finitely supported empirical distribution is one. Consequently the deterministic TV estimate must *not* be applied directly to empirical TV distance. Bounded moments provide a different route.

Proposition 12.1 (Elementary sampling upper bound). *Assume independent samples $Y_1, \dots, Y_N$ from μ_a , all supported on $[-R, R]$, and a known compact parameter neighborhood K as in (2.7). There is a constant C such that a feasible moment fit satisfies, with probability at least $1 - \beta$,*

$$d(\hat{a}, a) \leq C \left(\frac{\log(4r/\beta)}{N} \right)^{1/(2M(a^\circ))}, \quad (12.4)$$

for sufficiently large N . Without a known local neighborhood, the global bounded-box version gives exponent $1/(4r)$ in N .

Proof. For each $1 \leq n \leq 2r$, the sample average of Y_j^n has expectation $m_n(\mu_a)$. Normalize by R^n , so the summands lie in $[-1, 1]$. A bounded-variable exponential estimate gives

$$\Pr(|\tilde{m}_n - m_n| > R^n u) \leq 2e^{-Nu^2/2}.$$

For clarity, this follows by bounding the centered moment generating function of a variable in an interval of length two by $e^{\lambda^2/2}$, applying Markov's inequality to the sum, and optimizing at $\lambda = u$; apply the same argument to the negative sum. To justify the moment-generating-function bound, let $h(\lambda) = \log \mathbb{E} e^{\lambda(W - \mathbb{E}W)}$. Then $h(0) = h'(0) = 0$ and $h''(\lambda)$ is the variance under an exponential tilt. A distribution supported on an interval of length two has variance at most one, since its mean minimizes mean squared distance and its distance from the interval midpoint is at most one. Thus $h'' \leq 1$; integrating twice yields $h(\lambda) \leq \lambda^2/2$. A union bound over $2r$ moments and the choice $u = \sqrt{2 \log(4r/\beta)/N}$ give simultaneous moment bounds.

The finite cumulant polynomials are Lipschitz on the bounded set containing both the true and sample moment vectors. Dividing by the fixed nonzero γ_{2k} and subtracting the known background cumulants gives sum error $\eta \leq C_0 u$. Apply (12.3). The global statement follows from the global exponent $1/(2r)$ for finite-data error. $\square$

We do not claim that (12.4) is statistically minimax. A sharp statistical lower bound needs control of testing distances such as Hellinger distance or relative entropy, not merely the TV expansions between smooth densities established here. Endpoint flatness can matter to that stronger problem.

12.5 Rank is not continuously detectable at zero

Within capacity r , a zero factor can be replaced by an arbitrarily small positive one, changing the law by order t^2 while changing the number of positive factors by one. Thus the exact positive-factor count is not a continuous function of the observed law at a configuration with a zero slot. A positive lower scale bound or a thresholded notion of effective rank is necessary for robust rank reporting. This does not contradict exact recovery of the padded multiset from exact moments.

13 Verification record and reproducible computations

13.1 Exact checks

The accompanying `verification.py` was executed with Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. The exact checks use rational polynomial arithmetic. For every $m = 1, \dots, 12$, they verify the monic Chebyshev polynomial pair, equality of power sums through $m - 1$, the formula $D_m = m2^{1-m}$, the corresponding shifted-list equalities, simplicity of each polynomial, and disjointness of the two root sets. These are finite tests of identities already proved for all m in Lemma 7.1.

For capacities $r = 1, \dots, 8$, exact examples containing zero, repeated, and distinct rational squared scales were converted to output cumulants over the up background. The code then converts cumulants to raw moments, recovers cumulants from those moments, decodes the power sums, and reconstructs the monic polynomial. Each reconstructed polynomial equals the original exactly. For example, squared scales $(0, 3/5, 1/5, 1/5)$ give

$$P(z) = z^4 - z^3 + \frac{7}{25}z^2 - \frac{3}{125}z. \quad (13.1)$$

Sorting the scale list does not change this polynomial.

13.2 High-precision Fourier diagnostics

A separate numerical diagnostic evaluates characteristic-function differences at the single frequency $\xi = 0.9$. It uses 100 decimal digits of precision, Chebyshev clusters of orders 3 and 5, and $t = 2^{-\ell}$ for $\ell = 3, \dots, 12$. The fixed up multiplier is approximated by its first 220 sinc factors. At this frequency the omitted logarithmic tail is less than 10^{-132} in absolute value: for $|x| \leq 1/2$ one can bound

$|\log \operatorname{sinc} x| \leq x^2$, and the resulting tail is a geometric sum. This multiplier is common to the two configurations, so it does not control the observed cancellation order.

If $G(t)$ is the absolute Fourier difference, the diagnostic reports $\log_2(G(2t)/G(t))$. Its last computed values are:

Regime	Cluster size	Predicted power	Last halving slope
Positive collision	3	3	2.999999999868
Vanishing cluster	3	6	5.999999916416
Positive collision	5	5	4.999999999813
Vanishing cluster	5	10	9.999999863296

These are pointwise Fourier diagnostics, *not* numerical evaluations of total variation. They do not prove the analytical theorems, their L^1 remainders, or the optimality statements. Those conclusions follow from the proofs above. The explicit distinction prevents a cancellation plot at one frequency from being mistaken for a strong-metric convergence certificate.

13.3 How to reproduce and what was not checked

From the unpacked directory, run

```
python -m pip install -r requirements.txt
python verification.py --output-dir .
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The script writes `verification_results.json` and two CSV tables under `data/`. The assertions test exact identities and the stated numerical slopes. No infinite family is certified merely by passing a finite range, and the Python program is not part of a theorem-prover trusted boundary. No Lean or Rocq build was performed for this article. In particular, the repository’s existing formal identifiability result must not be interpreted as already formalizing the different stability theorems proved here.

14 A concrete formalization roadmap

A useful implementation should keep the algebraic and analytic boundaries separate. The algebraic core is smaller than the full probability-theoretic classification and admits independent testing.

Finite symmetric algebra. Formalize sorted nonnegative lists, squared-coordinate multisets, power sums, Newton reconstruction, and the coefficient recurrence. The initial endpoint is equality of multisets from the first r power sums over a characteristic-zero field. This should not depend on Fabius-specific analysis.

Quantitative root matching. Formalize an actual multiplicity-preserving bijection of roots, not a Hausdorff-set approximation. A complex-analytic route follows Lemma 5.1; a real-root-specific alternative might reduce dependencies. The isolated-factor lemma can then be proved by logarithmic contour integrals or an algebraic implicit-function argument at coprime factors. Neither route is represented here as already implemented.

Probability and finite cumulants. Connect the actual law of $B + \sum a_i U_i$ to finite raw moments and cumulants. Prove the uniform coefficient formula, nonvanishing, the known-background subtraction, and the explicit TV-to-moment bounds. The known up construction then becomes a specialization rather than part of every argument.

Smooth L^1 parameter dependence. The hardest analytic bridge is a reusable translation Taylor theorem in L^1 , followed by parameter differentiation of finite uniform convolutions. The Banach-valued symmetric Taylor argument must be connected to actual densities, not merely to formal characteristic-function series.

Two independent sharpness paths. Encode the Chebyshev level polynomials and their Newton sums. For positive collisions, identify a nonzero leading density coefficient using a compactly supported moment functional. For zero clusters, prove the exact coefficient in Theorem 10.1. These produce genuine counterexamples to stronger exponents. A proof of an upper bound alone is not a formalization of optimality.

Anchored positivity certificates. The polynomial Q in (11.1) gives a separate, comparatively elementary route to fixed-base stability. It is a good intermediate target: it captures the conceptual difference between anchored and pairwise inversion without requiring the entire Chebyshev family.

The desired final ledger should distinguish exact algebraic proofs, analytic bridges, imported background results, executable numerical tests, and remaining obligations. This article supplies theorem statements and conventional proofs, not untested code presented as machine-checked mathematics.

15 Further research questions and proposed projects

The following are proposed directions beyond the theorems proved here. Their status is a research agenda, not a verified claim that each is a previously published open problem.

Research question 15.1 (The optimal statistical experiment near collisions). For the same smooth-background model, determine the local minimax rate from N independent samples at each collision pattern. Does the rate $N^{-1/(2M)}$ from the moment upper bound hold sharply under natural conditions on the background? The concrete first target is to compute the Hellinger asymptotic of the Chebyshev pairs, including the contribution near the support boundary. An L^1 expansion alone does not justify squaring its leading term and dividing by a possibly flat density.

Research question 15.2 (The exact smoothing threshold). Replace C_c^∞ by finite Sobolev regularity, or remove the background altogether. Determine which collision patterns still have TV exponent $1/M$. The upper bounds already hold for any compactly supported known background. The missing issue is whether insufficient regularity exposes low-order moving knots before the moment-canceling mechanism reaches order m or $2m$.

Research question 15.3 (Unbounded capacity and tail-controlled infinite products). Let the unknown scale multiset be infinite with a prescribed summability or geometric-tail bound. Identify a metric on scale multisets and the sharp modulus of recovery. The finite-capacity exponent $1/(2r)$ degenerates with r , so no rank-independent positive power law can be inferred from this article. A useful first problem is to combine a certified tail truncation bound with the finite theorem and optimize the truncation depth as a function of noise.

Research question 15.4 (Unknown background parameters). Allow the background to belong to a known finite-dimensional family rather than being fixed. Determine when its cumulant directions can

be separated from the unknown uniform-factor directions. Merely subtracting an estimated background is not enough if the two parameter families share tangent or higher-order cumulant directions. A concrete target is a known smooth kernel with an additional unknown scale and a finite collection of uniform factors.

Research question 15.5 (Certified feasible fitting with sharp constants). Design an efficient certified algorithm for (12.2) or for its interval-data feasibility version. Quantify bit complexity in capacity, separation, requested amplitude precision, and distance to the nonnegative boundary. The theorem supplies optimal exponents, but the explicit general constants in Lemma 4.2 are deliberately coarse.

Research question 15.6 (Arithmetic scale restrictions). Study how the inverse problem changes when scales lie on a known dyadic or other discrete geometric grid. The repository’s analytic-order decoder suggests a different source of rigidity from unrestricted real-scale moment inversion. Determine noise thresholds for exact grid-scale recovery and compare zero-divisor measurements with finite cumulant measurements on the same model.

Research question 15.7 (Multivariate segment convolutions). Replace scalar amplitudes by unknown vectors and consider sums of uniform segments in $\mathbb{R}^d$ over a known smooth background. Even cumulants become symmetric tensors. Classify identifiability and stability modulo permutation and segment reversal, distinguishing collinear collisions from genuinely multidimensional degeneracies. The scalar Chebyshev obstruction embeds in every dimension, but need not exhaust the singularities.

Research question 15.8 (First complete formal classification). Formalize both Theorems 2.2 and 2.3, including explicit sharpness paths, in a proof assistant. An attractive staged project is first exact finite recovery, then anchored positivity certificates, then global root matching, and finally the smooth L^1 sharpness arguments.

16 Conclusion

The finite unknown-factor problem has a precise answer once its metric, capacity, background, and symmetries are fixed. Cumulants remove the known background and turn the unknown factors into a real-root polynomial. Positive root collisions determine one family of exponents; the square-root map at the nonnegative boundary doubles the zero-cluster multiplicity. Smoothness ensures that explicit moment cancellation survives in total variation, making the result sharp for actual probability laws.

The same analysis shows why the quantifiers matter. Uniform pairwise recovery near a singular configuration has exponent $1/M$, whereas recovery relative to that exact base has exponent $1/2$, except at the regular positive locus where it is Lipschitz. Confusing these two questions would obscure the strongest instability and simultaneously underestimate anchored stability.

The result is a finite-rank stability theorem with complete conventional proofs and reproducible algebraic checks. Infinite capacity, optimal statistical rates, weaker smoothing, and a full formal implementation remain separate research tasks rather than consequences asserted without proof.

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